

431 Quiz 1 INSTRUCTIONS ONLY for Fall 2026

Deadline: Wednesday 2026-10-14 at Noon

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2026-10-08

! Important

THIS PDF is a placeholder, containing just the instructions.
The real PDF will appear by 5 PM on Friday 2026-10-09.

Instructions for Students

There are **36** questions on this Quiz and this PDF is **36** pages long. Be sure you have all **36** pages¹. It is to your advantage to answer all **36** questions. Your score is based on correct responses and some partial credit. There's no chance a blank response will be correct, and a guess might be, so you should definitely answer all questions.

This is an open book, open notes quiz. You are welcome to consult the materials provided on the course website and that we've been reading in the class, but you are not allowed to post the questions online, use some sort of AI to help you, or discuss the questions on this quiz with anyone other than Dr. Love (not even the teaching assistants.) You will be required to complete a short affirmation that you have obeyed these rules as part of submitting the Quiz.

¹The last **3** pages are just session information.

0.1 The Google Form Answer Sheet

All of your answers should be placed in the Google Form Answer Sheet, located at...

<https://tinyurl.com/431-2026-quiz1-form>

All of your answers must be submitted through the Google Form by noon on Wednesday 2026-10-14, without exception. The form will close at 12:30 PM on that date, and no extensions will be available, so do not wait until late in the morning on Wednesday to submit your work. We will only accept responses through the Google Form.

The Google Form contains places to provide your responses to each question, and a final affirmation where you'll type in your name to tell us that you followed the rules for the Quiz. You must complete that affirmation before you can submit your responses. When you submit your results (in the same way you submit a Minute Paper) you will receive an email copy of your submission, with a link that will allow you to edit your work.

If you wish to work on some of the quiz and then return later, you can do this by [1] completing the final question (the affirmation) which asks you to type in your full name, and then [2] submitting the quiz. You will then receive a link at your CWRU email which will allow you to return to the quiz as often as you like without losing your progress.

0.2 The Data Sets

I have provided **six** data sets (called **q01.xlsx**, **q14.xlsx**, **q16a.csv**, **q16b.csv**, **q23.csv** and **q35.csv**) that are mentioned in the Quiz. These can either be downloaded from our 431-data page, or found in our Shared Drive in the Quiz 1 folder's data sub-folder.

0.3 Getting Help

If you need clarification on a Quiz question, you have exactly two ways of getting help:

- Ask your quiz question via email to **431-help at case dot edu**.
- Ask your quiz question in person during [Class 15](#) on 2026-10-13.

During the Quiz period (5 PM 2026-10-09 through 12:30 PM 2026-10-14) we will not answer questions about the Quiz in TA office hours, or in any other way. Instead, I will respond to questions sent before 9 AM on 2026-10-14 in a timely fashion. Some cautions:

- Specific questions are more likely to get helpful answers.
- I will not review your code or your English for you.
- I will not tell you if your answer is correct, or if it is complete.
- I will email all students if we find an error in the Quiz that needs fixing.

0.4 When Should I Ask For Help?

We recommend the following process.

- If you encounter a tough question, skip it, and build up your confidence by tackling other questions.
- When you return to the tough question, spend no more than 10-15 minutes on it. If you still don't have it, take a break (not just to do other questions) but an actual break.
- When you return to the question, it may be much clearer to you. If so, great. If not, spend 5-10 minutes on it, at most, and if you are still stuck, ask Dr. Love a question via email.
- This is not to say that you cannot ask us sooner than this, but you should **never, ever** spend more than 20 minutes on any question without asking for help.

If you are stuck, ASK A QUESTION

You should **NEVER** spend more than 20 minutes on any question without asking Dr. Love for help. Just email him at **431-help AT CASE DOT EDU²**.

0.5 Scoring the Quiz

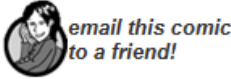
The questions are worth **3** points each yielding a total of **108** points. The questions are not in any particular order, and range in difficulty from “things I expect everyone to get right” to “things that are deliberately tricky”.

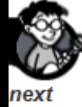
²If you ignore this advice, that's your decision, but you'll have to live with the consequences. If you're taking more than 20 minutes to answer any of these questions, then you're very likely to be misunderstanding something.

0.6 How long should this take?

The Quiz is meant to take 6-7 hours. I expect most students will take 4-9 hours, and some will take as little as 3 or as many as 10. Some questions will take more time than others.

On the other hand...





How long your Prof. thinks it should take to do something		How long it'll actually take you to do it
↓		↓
"Trivial"	=	There goes your week.
"Easy enough"	=	Pull your hair out for a month.
"About a week"	=	Actually, this is pretty easy. He/she doesn't know there's technology that will do this for you now. Take the week off!
"Should keep you occupied for the rest of the term"	=	He/she will forget they asked you to do this by the end of the term. Don't even bother.
"This might make a good thesis topic"	=	Say hello to your thesis topic.
"Hmmm..."	=	Uh oh.





WWW.PHDCOMICS.COM

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Source: <https://phdcomics.com/comics/archive.php?comid=1093>

0.7 Writing Code into the Google Form

Occasionally, we ask you to provide a single line of code. If not otherwise specified, a single line of code in response can contain **at most** two pipes, although you may or may not need the pipe in any particular setting. Note that I exclusively used the `|>` pipe, and not the `%>%` pipe, in developing this Quiz, but you may use either in your responses.

Do not include the `library` command at any time for your responses on the Quiz that ask for code. Assume in all questions that all packages listed below are loaded in R.

0.8 R Packages and Love-431.R script

Here is a list of packages in R. This **doesn't mean you need to use all of these packages**, and it also doesn't mean that I actually used all of them in building the Quiz and its answer sketch. It means only that I didn't use any other packages, other than **xfun** for session information, so you can do the entire Quiz without using any other packages.

I have also provided this listing to you at [this link](#), so you can easily copy and paste it into your own RStudio session.

```
knitr::opts_chunk$set(comment = NA)

library(here)
source(here("data/Love-431.R"))

library(car)
library(DescTools)
library(Epi)
library(glue)
library(infer)
library(janitor)
library(knitr)
library(MKinfer)
library(naniar)
library(palmerpenguins)
library(patchwork)
library(readxl)
library(rstanarm)
library(easystats)
library(tidyverse)

theme_set(theme_bw())
```